

Data Transformation & Data Cleaning using Power Query

A training handout covering core cleaning and shaping techniques inside the Power Query Editor.

1. Why Clean & Transform Data?

Raw data from source systems is rarely analysis-ready — it may contain blanks, duplicates, inconsistent formats, wrong data types, or unnecessary columns. **Power Query** is Power BI's built-in engine for cleaning and reshaping data before it reaches the report/data model. Every action is recorded step-by-step in the **Applied Steps** pane, making the process transparent, repeatable, and refreshable.

Access it via: **Home** → **Transform Data** (Power BI Desktop) or **Data** → **Get Data** → **Transform Data** (Excel).

2. Data Cleaning Techniques

2.1 Handling Rows & Columns

- **Remove Columns** — right-click a column header → Remove, to drop unused fields.
- **Remove Duplicates** — select column(s) → Remove Rows → Remove Duplicates.
- **Remove Errors** — Remove Rows → Remove Errors, to drop rows that fail a transformation.
- **Remove Blank Rows** — Remove Rows → Remove Blank Rows.
- **Keep/Remove Top or Bottom Rows** — useful for skipping report headers/footers in messy exports.
- **Filter Rows** — use column dropdown filters or Text/Number/Date Filters for conditional cleaning.

2.2 Fixing Data Quality Issues

Issue	Power Query Fix
Inconsistent casing (e.g. "delhi" vs "Delhi")	Transform → Format → UPPERCASE / lowercase / Capitalize Each Word
Extra spaces / hidden characters	Transform → Format → Trim, Clean
Blank cells that should repeat a value	Transform → Fill → Down (or Up)
Wrong or mixed data types	Right-click column → Change Type (Text, Whole Number, Decimal, Date, etc.)

Null / error values	Replace Values, or Remove Errors / Remove Rows with nulls
Inconsistent category names (e.g. "NY" vs "New York")	Replace Values, or Transform → Replace Values

2.3 Replace Values & Conditional Cleaning

Use **Transform** → **Replace Values** to swap specific text/numbers, or **Add Column** → **Conditional Column** to create a new cleaned field based on if/then logic without writing formulas manually.

3. Data Transformation Techniques

3.1 Reshaping Columns

- **Split Column** — by Delimiter, Number of Characters, or Positions (e.g. split "First Last" into two columns).
- **Merge Columns** — combine multiple columns into one (e.g. First + Last → Full Name).
- **Extract** — pull Text Before/After Delimiter, Length, First/Last Characters, or Range.
- **Pivot Column** — turn row values into new columns (wide format) — useful for summarizing categories.
- **Unpivot Column(s)** — turn columns into rows (long format) — essential for tidy, report-ready data.
- **Transpose** — swap rows and columns entirely.

3.2 Combining Data

Operation	Use When	Equivalent To
Merge Queries	Combining columns from two tables using a common key	SQL JOIN (Inner/Left/Right/Full/Anti)
Append Queries	Stacking rows from tables with the same structure	SQL UNION

Merge Queries join types available: Inner, Left Outer, Right Outer, Full Outer, Left Anti, Right Anti — chosen from the Merge dialog's join kind dropdown.

3.3 Grouping & Aggregating

- **Group By** (Transform tab) — aggregate rows by one or more columns: Sum, Average, Count, Min, Max, Count Distinct Rows.
- Use **Add grouping** and **Add aggregation** to build multi-level summaries directly in Power Query.

3.4 Adding Calculated Columns

- **Add Column** → **Custom Column** — write an M formula to create a new field (e.g. Profit = [Sales]-[Cost]).
- **Add Column** → **Conditional Column** — build if/then/else logic without code, via a guided UI.
- **Add Column** → **Index Column** — add a running row number, useful as a surrogate key.
- **Add Column** → **Column From Examples** — Power Query infers the transformation logic from sample output you type in.

3.5 Data Type Management

Correct data types (Text, Whole Number, Decimal Number, Date, Date/Time, True/False) are essential — they affect available transformations, DAX calculations, and report visuals. Set types early using the column header type icon or right-click → Change Type, then **Locale** for correct date/number parsing.

Example M code combining several steps:

```

let
    Source = Excel.CurrentWorkbook(){[Name="RawSales"]}[Content],
    RemovedCols = Table.RemoveColumns(Source, {"Comments"}),
    TrimmedText = Table.TransformColumns(RemovedCols, {"City",
Text.Trim})),
    FilledDown = Table.FillDown(TrimmedText, {"Region"}),
    ChangedType = Table.TransformColumnTypes(FilledDown, {"OrderDate", type
date})),
    RemovedDup = Table.Distinct(ChangedType)
in
    RemovedDup

```

4. Working with Applied Steps

- Every click generates a new step, listed top-to-bottom in the **Applied Steps** pane.
- Click the gear icon next to a step to edit its settings without redoing it.
- Right-click any step → **Rename** to give it a meaningful name for documentation.
- Steps can be reordered by drag-and-drop, but later steps must still be compatible with earlier changes.
- Use **Insert Step After** to add a transformation in the middle of the sequence when needed.

5. Best Practices for Cleaning & Transforming

- Clean and filter data **as early as possible** in the step sequence — smaller data transforms faster.
- Set correct **data types immediately after extraction**, before other transformations depend on them.
- Prefer **Group By** and **Merge** over manual formulas where possible — they're easier to audit.
- Keep transformations **query-foldable** (filters, renames, type changes) when connected to a database, so Power BI pushes the work back to the source for speed.
- Name and organize steps clearly — future you (or your students) will thank you during refresh troubleshooting.
- Avoid loading unnecessary columns — trim the dataset to only what the report needs.

6. Quick Reference Cheat Sheet

Task	Menu Path
Remove duplicates	Home → Remove Rows → Remove Duplicates
Trim/Clean text	Transform → Format → Trim / Clean
Change data type	Column header icon, or right-click → Change Type
Fill blanks	Transform → Fill → Down/Up
Split a column	Transform → Split Column

Merge two tables	Home → Combine → Merge Queries
Append tables	Home → Combine → Append Queries
Group and summarize	Transform → Group By
Pivot / Unpivot	Transform → Pivot Column / Unpivot Column(s)
Add calculated column	Add Column → Custom Column
Conditional logic column	Add Column → Conditional Column

Prepared as a training handout for data analytics learners.